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Adapting a Fetal Brain Super-Resolution Quality Control Framework for Eye MRI

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Preface

Abstract

Automated quality control (QC) is crucial for reliable super-resolution reconstructed (SRR) ocular MRI. This project successfully adapted the FetMRQC_SR framework, initially for fetal brain MRI, to create a robust automated QC pipeline for SRR ocular imaging. Key achievements include significantly enhancing Image Quality Measures (IQMs) by adapting existing brain metrics and introducing novel eye-specific morphological IQMs, such as globe sphericity and lens aspect ratio. Additionally, MRIQC-inspired metrics and comprehensive segmentation-based statistics were integrated.

The adapted MREyeQC_PA system now efficiently extracts quantitative features from ocular MRI data, overcoming technical challenges. An initial Random Forest classifier was trained using these IQMs and binarized manual QC ratings. The workflow included feature scaling, hyperparameter tuning via stratified 5-fold cross-validation, and class imbalance correction strategies. Preliminary evaluation on a limited test set ($N = 20, 15$

This work establishes a critical foundation and comprehensive IQM framework for objective automated QC in SRR ocular imaging research. While the initial model underscores the need for dataset expansion, especially for underrepresented quality classes, and further model refinement, the developed pipeline offers a robust platform for future advancements to enhance the reliability and reproducibility of quantitative ophthalmic MRI studies. **Keywords:** Eye MRI, Super-

Resolution, Quality Control, Image Quality Metrics, Segmentation, Topology, Foreground Masking, Machine Learning, Cross-Validation

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1 | Introduction

Magnetic resonance imaging (MRI) has consolidated its role as a powerful, non-invasive tool for visualizing the complex anatomy of the human eye and surrounding orbital structures [see [1, 2]]. Its applications in ophthalmology continue to expand, offering crucial insights into the diagnosis, monitoring, and understanding of a wide range of ocular pathologies [see [3, 4]]. The advent of MRI techniques promises to offer even finer anatomical details. This additional precision is essential for the assessment of subtle structural changes and for advancing research in ocular physiology and pathophysiology, including functional studies such as retinal fMRI and vascular oxygenation research [see [5, 6]].

However, the diagnostic and experimental utility of ocular MRI relies critically on obtaining high-quality images. The unique characteristics of the eye (small size, mobility, and complex layered structures) present significant imaging challenges [see [2, 3]]. Image quality can be degraded by various artifacts, including those related to involuntary saccades and blinks, as well as physiological movements such as blood flow and cerebrospinal fluid pulsation [see [7]]. Further complications arise from the inherent complexities of MRI acquisition, such as susceptibility artifacts near air-tissue interfaces and tradeoffs between spatial resolution, signal-to-noise ratio (SNR), and acquisition time [see [4]]. These factors can obscure critical diagnostic information, lead to misinterpretation, or reduce the reliability of quantitative measurements [see [8] for an example of accurate measurements]. Therefore, quality control (QC) methods and objectives for ocular MRI data are essential to ensure data fidelity and reliability. Although automated quality control frameworks have seen considerable development and adoption for other anatomical regions, including neuroimaging (MRIQC for brain MRI [9], which forms the basis of FetMRQC_SR) [10]. Existing quality control tools are often not directly transferable or optimal for ocular imaging due to specific anatomical features, distinct image contrast properties, and common artifact profiles. Manual quality control, involving visual inspection by trained experts, remains a common practice. However, manual assessment is inherently subjective, labor-intensive, and subject to inter- and intra-rater variability, making it a bottleneck for large-scale clinical studies, longitudinal research, and standardized multicenter trials.

This project, MREyeQC_PA, directly addresses this gap. It aims to improve and adapt the FetMRQC_SR framework, originally designed for fetal brain MRI quality control ([10]), to establish a robust and automated quality control pipeline specifically tailored for SRR ocular MRI. The main objective is to develop a system capable of quantitatively assessing the quality of ocular MR images through the calculation of a relevant suite of image quality measures (IQMs). These IQMs are designed to encompass a wide range of quality aspects, including general image properties (e.g., noise, sharpness, contrast), morphological features of key ocular structures (such as the lens and globe), and indicators of common imaging artifacts. Subsequently, these quantitative measurements are used to train machine learning models aimed at predicting image quality with accuracy and consistency comparable to, or even superior to, that of human experts.

2 | IQM Selection Strategy: Linking Ratings to Image Features

Identifying the most effective image quality indicators (IQMs) for a machine learning model to differentiate between good and poor quality images is a crucial step in this project. This step relies on the analysis of manually assessed images and their associated artifact scores to highlight correlations between visual features and perceived quality. This chapter first details the reliability of the manual scoring process by analyzing the consistency of human assessments by raters. It then discusses the influence of various image artifacts on these human quality judgments. Finally, it describes the strategic selection of IQMs, informed by both statistical analysis and visual features observed in different image quality classes.

2.1 Analysis of Artifact Influence on Image Quality Ratings

To effectively select relevant IQMs to train a good and bad quality classification model, we must first understand what factors determine image quality. To do this, there are two key steps

1. **Ensuring Inter-Rater Reliability:** Confirm the consistency of the evaluators' assessments. It is necessary to ensure that they are able to distinguish a good image from a bad one.
2. **Identifying Impactful Artifacts:** Determine the specific image artifacts that most impacted their overall quality judgments. This allows IQMs to be prioritized to the most critical types of degradation.

2.1.1 Establishing Inter-Rater Reliability: How Consistent Are The Raters?

Before using manual image ratings as a benchmark for IQM selection, it's essential to confirm that our three expert raters applied the quality criteria consistently. If their agreement is low, the ratings are less reliable as a "ground truth." This section assesses this consistency using several standard statistical measures and provides an initial overview of rating distributions.

Initial Rater Distribution Analysis

First, a pie chart was generated to visualize the distribution of ratings assigned by each expert rater. This provides an initial insight into how frequently each rater assigned images to different quality categories, hinting at potential inter-rater variability before formal statistical assessment

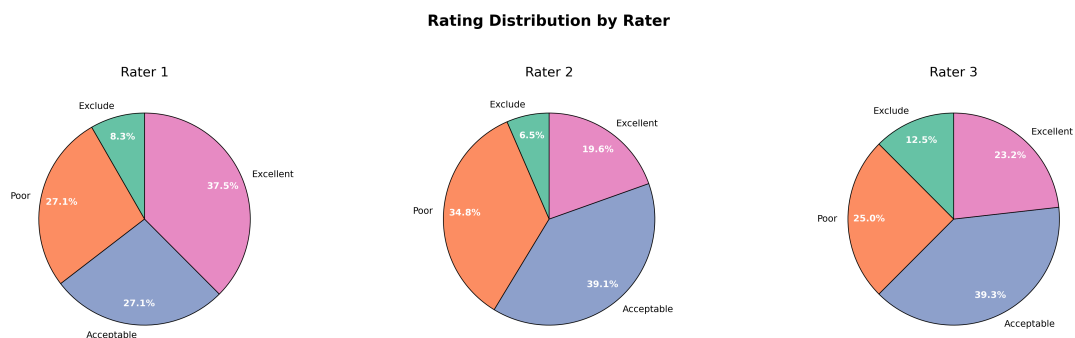


Figure 2.1 Rating distribution by rater across 'Exclude', 'Poor', 'Acceptable', and 'Excellent' categories. Note the variations in how frequently each rater assigned images to the different quality categories. Figure from others degradation are in A

As shown in Figure 2.1, analysis of image quality ratings reveals significant inter-rater variability across the four defined categories: “Exclude,” “Poor,” “Acceptable,” and “Excellent.” This demonstrates, first of all, the difficulty of manually classifying image quality. This already represents a challenge for the project. A key observation is the varying rigor between raters. Rater 3, for example, displayed the strictest rating trends, classifying 12.5% of images as “Exclude.” This figure is significantly higher than that of Rater 1 (8.3%) and is nearly double the proportion designated by Rater 2 (6.5%) for the same critical category. Conversely, Reviewer 1 showed the greatest tolerance for good quality images, rating 37.5% of images as “Excellent,” compared to 19.6% for Reviewer 2 and 23.2% for Reviewer 3.

Overall, differences are discernible in how each reviewer interprets the quality assessment criteria. This variability may pose a challenge when training the model

Inter-Rater Reliability Assessment Methodology

To quantitatively assess the consistency between the three raters, 2 statistical measures are calculated for each image quality measure assessed:

- **Intraclass Correlation Coefficient (ICC):** Specifically, the ICC(A,k) coefficient is used to assess the agreement of the average measurements of $k = 3$ raters. This coefficient indicates the consistency with which raters rated the images on a continuous scale (0-4). Values close to 1 indicate higher agreement, with an ICC coefficient > 0.75 generally considered good to excellent reliability. **add reference**
- **Percentage Agreement:** For the 'Eyes Open Status', a simple percentage agreement was also determined, representing how often all three raters assigned the exact same category.

Inter-Rater Reliability Assessment Results

The results of this reliability analysis are summarized in Table 2.1 and further illustrated by the correlation matrices in Figures 2.2.

Table 2.1 *Inter-Rater Reliability Metrics for Image Quality Assessments*

Metric	ICC(A,k)	ICC CI95% Low	ICC CI95% High	% Agreement
rating	-0.149	-1.490	0.520	-
rating1	0.144	-0.820	0.640	-
rating2	-0.293	-1.800	0.460	-
blur	-0.289	-1.510	0.430	-
noise	-0.404	-1.960	0.400	-
motion	0.136	-0.800	0.630	-
bgair	-0.328	-2.020	0.450	-
Eyes Open Status	-	-	-	40.000

The intraclass correlation coefficient values for overall quality assessments (“rating”, “rating1”, “rating2”) and specific artifact scores (“blur”, “noise”, “motion”, “bgair”) are consistently low, often negative. For example, “rating” yielded an ICC(A,k) of -0.149 (95% CI: [-1.490, 0.520]), and “noise” an ICC of -0.404 (95% CI: [-1.960, 0.400]). According to common interpretation guidelines ([11]), ICC values below 0.5 indicate low reliability. The resulting ICC values, many of which are negative or close to zero, and their wide 95% confidence intervals, suggest a lack of consistency among raters when assessing these measures on a continuous scale. For the "Eyes Open State" measure, the percentage agreement is 40.00%. This relatively low agreement further highlights the difficulty of obtaining consistent manual assessments.

Overall, the inter-rater reliability analysis consistently indicates low consistency between the three raters for the majority of image quality indicators assessed. This low reliability suggests the difficulty of systematically applying the evaluation criteria and the difficulty of having an accurate ground truth.

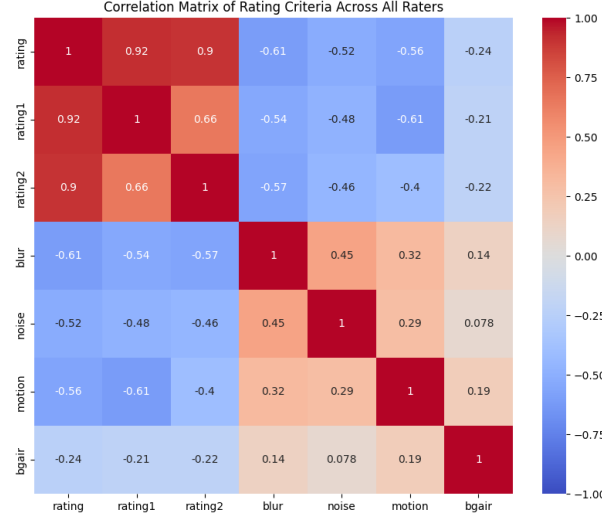


Figure 2.2 Correlation Matrix of rating criteria across all raters

Impact of Blur on Overall Rating: The artifact exhibiting the strongest correlation with our aggregate image quality rating is blur, with a coefficient of -0.61. This indicates a substantial inverse relationship: as the perceived blur increases, the overall image quality rating tends to decrease significantly.

Inter-Artifact Correlation (Blur and Noise): A moderate positive correlation of 0.45 is observed between blur and noise. This suggests that images high in blur are often also perceived as noisy, potentially reflecting common degradation pathways or the challenge in distinguishing these artifacts during human assessment.

2.1.2 Which Image Artifacts Most Impact Human Quality Judgments?

After assessing the reliability (and limitations) of the manual assessments, the next objective was to identify the specific visual artifacts with the most significant independent influence on overall image quality assessments. Understanding this is essential for prioritizing IQMs sensitive to the most significant image degradations. To achieve this, a Linear Mixed-Effects Model (LMM) was employed. This statistical approach allows for the modeling of the relationship between the overall quality score and the scores of individual artifacts.

Key Findings from the LMM:

The LMM results (summarized in Table 2.2) pinpointed the artifacts that statistically significantly degraded the perceived image quality:

- **Blur:** Showed the **strongest negative impact**. For each one-unit increase in perceived blur, the overall quality rating decreased by approximately 0.288 ($p < 0.001$).
- **Noise:** Also a significant factor, with each unit increase in noise lowering the quality rating by about 0.254 ($p < 0.001$).
- **Motion:** Had a significant negative effect, reducing the quality rating by roughly 0.204 ($p = 0.001$).

- **Background Artifacts ('bgair'):** In this model, once blur, noise, and motion were accounted for, 'bgair' did not show a statistically significant independent effect on the overall quality rating ($p = 0.089$). While not statistically significant in this specific model, visual reports and qualitative analysis still suggested 'bgair' as a relevant factor.

Implications for IQM Selection:

The LMM analysis provides clear guidelines for the IQM strategy: Blur, noise, and motion are the main culprits of image quality degradation from the rater's perspective, with blur being the most dominant. Therefore, IQMs must include robust metrics capable of detecting and quantifying these three artifacts. While background artifacts may still be relevant (as suggested by the visual reports and the initial pie chart analysis, Section 2.1.1), their direct and independent impact on the overall rating appears less critical than the other three when modeled together.

Table 2.2 *Simplified Linear Mixed-Effects Model Results (Dependent Variable: rating)*

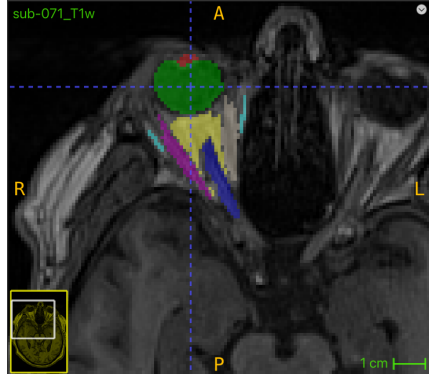
Dependent Variable: rating						
Effect	Coef.	Std.Err.	z	P> z	[0.025	0.975]
Intercept	3.081	0.106	29.159	0.000	2.874	3.288
blur	-0.288	0.064	-4.522	0.000	-0.412	-0.163
noise	-0.254	0.070	-3.645	0.000	-0.391	-0.118
motion	-0.204	0.061	-3.313	0.001	-0.324	-0.083
bgair	-0.122	0.072	-1.701	0.089	-0.263	0.019
Random Effect Variance:						
sub Var	0.075	(SE: 0.098)				
Model Fit:						
No. Obs: 151, No. Groups: 83, Scale (Residual Var): 0.1809, Log-Likelihood: -115.38						
Converged: Yes, Method: REML						

2.2 Visual Analysis of Image Quality Classes

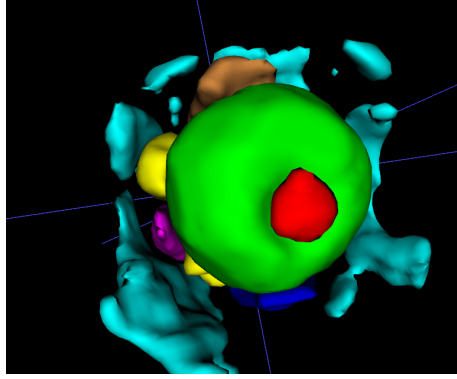
In order to refine the selection of IQMs and to complete the statistical results, a qualitative visual evaluation is carried out on images manually classified into three categories: "Excluded", "Poor" and "Excellent". This observation is made on the segmentation carried out for a pre-trained nnUnet v1 model and 3D images from eye in order to see on which class the model has difficulty segmenting.

2.2.1 Excluded Images: Severe Degradation

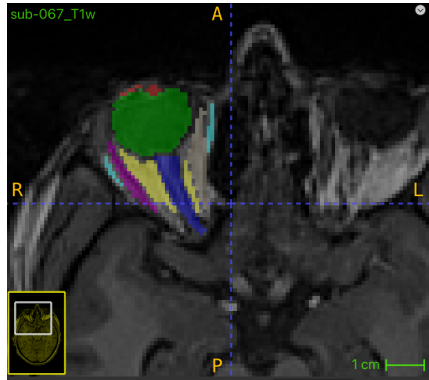
Excluded images are characterized by severe degradation, rendering them unusable for analysis.



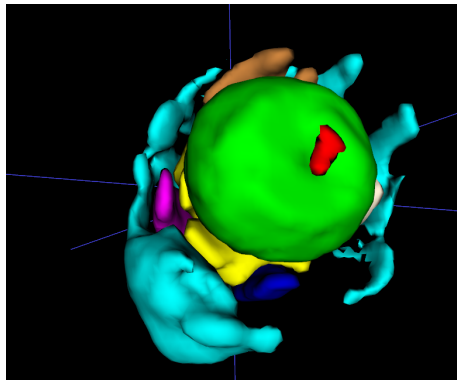
(a) *T1w - Subject 071*



(b) *T1w_3D - Subject 071*



(c) *T1w - Subject 067*



(d) *T1w_3D - Subject 067*

Figure 2.3 Comparison between T1w (left column) and T1w_3D (right column) images categorized as 'Excluded' quality. These examples demonstrate severe lens segmentation issues, geometric distortions, and general image degradation.

Observation

In the segmented excluded images, severe deficiencies are consistently observed. The lenses are poorly segmented, often deviating significantly from their characteristic ellipsoidal shape. Similarly, the globe segmentation exhibits notable surface imperfections. Furthermore, muscle segmentation frequently contains aberrations, with multiple unrelated pieces being segmented, leading to overall poor quality.

2.2.2 Bad Quality Images: Noticeable Imperfections

Images in the "Bad Quality" (poor and acceptable class) category are not necessarily unusable but show noticeable imperfections that distinguish them from excellent quality, yet they are not as severely compromised as "Excluded" images.

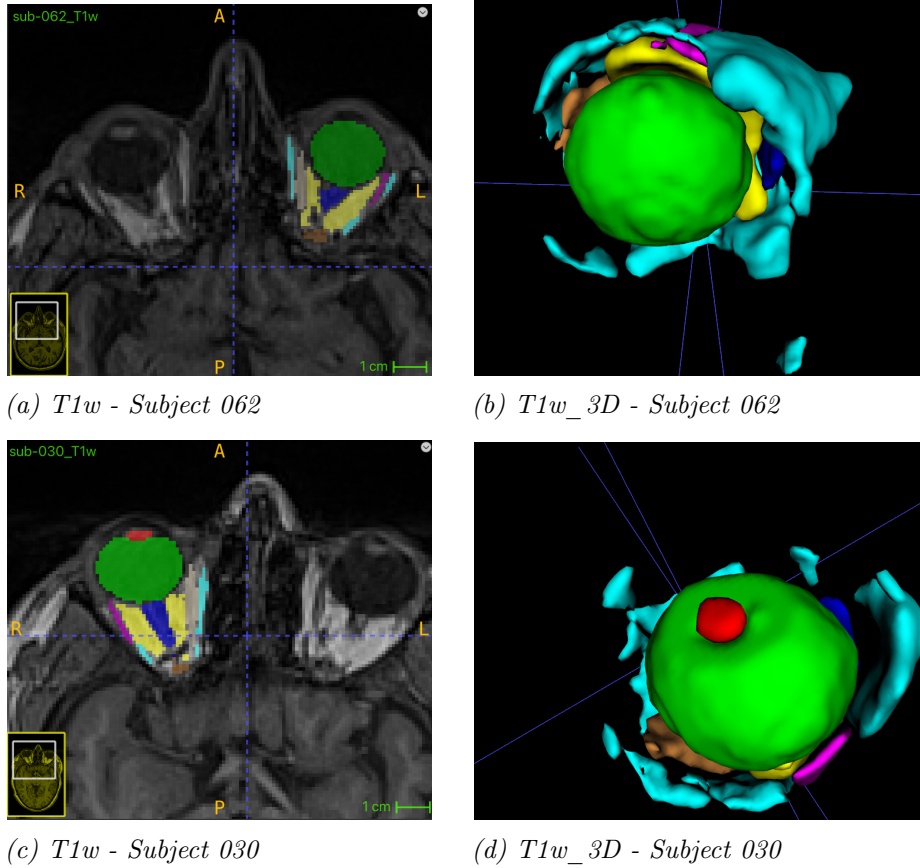


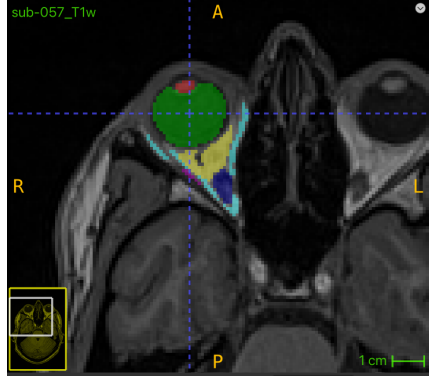
Figure 2.4 Comparison between *T1w* (left column) and *T1w_3D* (right column) images categorized as 'Bad' quality. These examples show noticeable but not severe blur and minor segmentation imperfections.

Observation

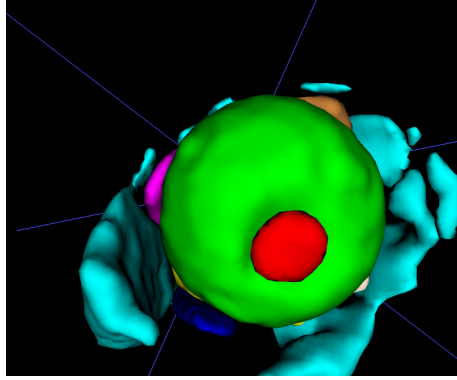
The 'Poor Quality' category, while also exhibiting aberrations in lens, globe, and muscle segmentations, differs from the 'Excluded' category in that these imperfections are typically isolated and less severe. Consequently, images in this category are generally still considered usable

2.2.3 Excellent Quality Images: High Fidelity

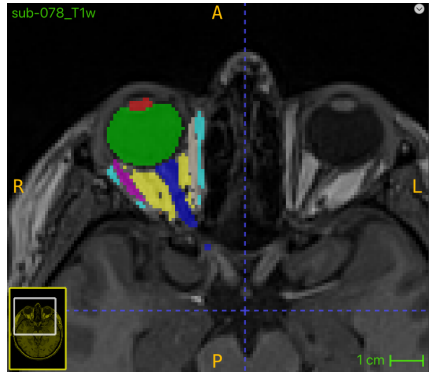
Excellent quality images represent the ideal standard, characterized by high fidelity and clarity, making them optimal for subsequent analysis.



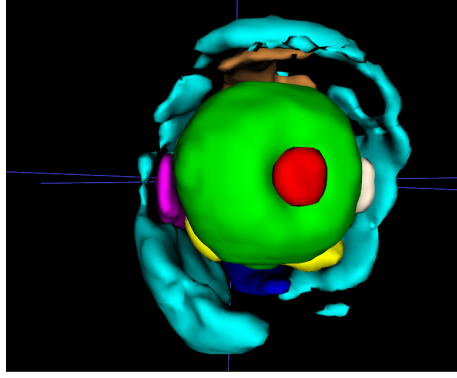
(a) T1w - Subject 057



(b) T1w_3D - Subject 057



(c) T1w - Subject 078



(d) T1w_3D - Subject 078

Figure 2.5 Comparison between T1w (left column) and T1w_3D (right column) images categorized as 'Excellent' quality. These examples showcase clear anatomical delineation, accurate segmentations, and minimal artifacts.

Observation

Images categorized within this class are distinguished by segmentations of exceptionally high quality. Their surfaces are precisely delineated, and the morphology of the various ocular components faithfully adheres to established human anatomy. This class is notably characterized by the near-complete absence of image deterioration

2.2.4 Observations Summary for IQM Selection

In the segmented excluded images, the lenses exhibit distinct segmentation problems. These lenses often display discontinuous or aberrant shapes that do not correlate with the true anatomical form of a human lens, underscoring the importance of incorporating lens-segmentation-based IQMs.

Regarding the Globe, it is observable that globes in excellent quality images possess a notably smoother surface compared to those found in excluded or poor-quality images. This distinction highlights the importance of basing IQMs on this segmentation class as well.

For the other classes, no noticeable differences are consistently apparent between excluded images and excellent quality ones. This observation is attributed to the inherent complex morphology of these classes, which impedes the easy visual identification of segmentation discrepancies.

3 | Methodology: Adapting FetMRQC_SR for Eye Tissues and ML Workflow

With a comprehensive understanding of the image data and existing segmentations, the next phase involved focusing on adapting and developing Image Quality Metrics (IQMs) specifically for the different ocular tissues. For this purpose, FetMRQC_SR [10] was utilized, a specialized subcode of the MRIQC project [9] originally developed for fetal brain imaging

3.1 Segmentation Adaptation for Eye Anatomy

One of the primary tasks was to adapt the existing project framework to leverage anatomical segmentation of the eye. This integration is paramount, as these segmentations provide the essential anatomical context upon which all subsequent image quality metric calculations will be based.

3.1.1 Defining Target Eye Tissues

To allow the calculation of IQMs on specific anatomical regions of the eye, these parts first needed to be precisely defined. They were categorized into the following classes: 'LENS', 'GLOBE', 'OPTIC NERVE', 'FAT', 'MUSCLE', and 'BGAIR'. These definitions directly correspond to the segmentation groups that will be utilized for IQM computation.

3.2 Image Quality Metric (IQM) Adaptation and Expansion

With these anatomical classes now defined, the adaptation of existing IQMs within the FetMRQC_SR project was undertaken. Additionally, novel metrics, specifically the sphericity index and the aspect ratio (ratio of major and minor axes) of the lens, were developed to further quantify image quality.

Categorization of IQM Adaptations

Adapting IQMs can be categorized into several functions.

1. **Direct Application of General-Purpose IQMs:** Metrics that assess global image characteristics without relying on specific anatomical priors.
2. **Re-targeting of Brain-Specific IQMs:** Metrics originally defined for specific brain tissues (e.g., Gray Matter, White Matter) that required re-definition for analogous or relevant ocular structures (e.g., Lens, Globe).
3. **Development of Novel Eye-Specific IQMs:** New metrics designed to capture unique morphological or quality aspects of ocular anatomy.
4. **Systematic Application of Segmentation-Based Metrics:** Extension of existing statistical and topological metrics to all newly defined ocular tissues.

The following sections elaborate on each of these categories. For a comprehensive list of all IQMs integrated into the MREyeQC_PA framework, please refer to Appendix B

3.3 Direct Application of General-Purpose IQMs

Image Quality Metrics (IQMs) designed to assess general image properties can be directly applied without modification, as they utilize the global image rather than requiring specific segmentations. These metrics are effective in characterizing overall image intensity distribution, noise profiles, information content, inter-slice coherence, and fundamental volumetric properties of the primary region of interest.

These retained and directly applied IQMs include:

- **Intensity Statistics:** mean, median, std (standard deviation), variation (coefficient of variation), kurtosis, percentile_5, percentile_95.
- **Volumetric and Positional Properties:** mask_volume, centroid, centroid_full.
- **Information-Theoretic Measures:** shannon_entropy, joint_entropy_median, mi_window (Mutual Information), nmi_window / nmi_median (Normalized Mutual Information).
- **Inter-Slice/Window Consistency & Quality:** ncc_window / ncc_median (Normalized Cross-Correlation), psnr_window (Peak Signal-to-Noise Ratio), rmse_window / nrmse_window (Root Mean Squared Error), mae_window / nmae_window (Mean Absolute Error), ssim_window (Structural Similarity Index).
- **Filter-Based Edge/Texture Metrics:** filter_laplace, filter_sobel.
- **Other general metrics:** rank_error.

3.4 Re-targeting of Brain-Specific IQMs for Ocular Anatomy

Certain IQMs within the FetMRQC_SR project originally rely on specific segmentations, such as gray matter (GM) and white matter (WM). To ensure their relevance for ocular MRI, their fundamental principles have been preserved, but their application has been reoriented to target key ocular structures instead

- **Contrast-to-Noise Ratio (CNR) and Coefficient of Joint Variation (CJV):**
 - *Original Context (Brain):* These metrics assessed the contrast and signal homogeneity/variability between GM and WM.
 - *Ocular Adaptation:* For eye imaging, these metrics is adapted to evaluate the relationship between the **LENS** and the surrounding **GLOBE** tissue. This involved modifying the metric computation to use the intensity statistics derived from the segmented LENS and GLOBE masks. The aim is to quantify the conspicuity of the lens within the globe and the homogeneity of these structures.
- **Tissue-to-Maximum-Intensity Ratio (adapted from WM2MAX):**
 - *Original Context (Brain):* wm2max measured the ratio of the mean white matter intensity to a high percentile of the total image intensity, often an indicator of signal saturation or abnormal hyperintensities.

- *Ocular Adaptation*: This concept is applied individually to both the **LENS** and the **GLOBE**. It calculates the ratio of the mean intensity within the specified ocular structure to the 99.95th percentile of intensity values across the entire processed eye image. This can help detect issues like unusually low signal within these structures or global image intensity artifacts.

- **Signal-to-Noise Ratio - Dietrich's Method (SNRd)**:
 - *Original Context (Brain)*: Calculated using the mean of a foreground tissue (e.g., GM) and noise statistics from an "air" mask outside the head.
 - *Ocular Adaptation*: Applied to both **LENS** and **GLOBE** as foreground regions and with the **BGAIR**.

- **Foreground-Background Energy Ratio (FBER)**:
 - *Original Context (Brain)*: Ratio of mean energy within the head mask to energy outside (background air).
 - *Ocular Adaptation*: Applied to both **LENS** and **GLOBE** as foreground regions and with the **BGAIR**.

- **Entropy Focus Criterion (EFC)**:
 - *Original Context (Brain)*: A global image metric assessing ghosting and blurring from the entropy of all image intensities (excluding pure background frame).
 - *Ocular Adaptation*: Calculated on the entire preprocessed (cropped and right-eye filtered) eye image. The interpretation remains similar: lower values indicate sharper, less dispersed energy, implying better quality.

- **Residual Partial Volume Effect (RPVE)**:
 - *Original Context (Brain)*: Typically calculated for interfaces between major tissues like GM-WM or GM-CSF, representing the intensity difference or deviation at tissue boundaries.
 - *Ocular Adaptation*: Implemented to assess interfaces relevant to the eye:
 - * **rpve_lens_boundary**: Evaluates the intensity profile at the boundary of the LENS (interface with the surrounding GLOBE content).
 - * **rpve_globe_boundary**: Evaluates the outer boundary of the GLOBE with surrounding orbital tissues.
 - * **rpve_globe_lens_interface**: Specifically focuses on the interface region between the GLOBE's internal structures and the LENS surface.

3.4.1 New Eye-Specific Morphological IQMs

Consistent with the discussions in Chapter 2, it was determined that certain anatomical regions of the eye would benefit from specialized quality metrics. Consequently, the sphericity index and the lens aspect ratio were developed for this purpose. The following section details their capabilities regarding image quality:

Globe Sphericity (`seg_globe_sphericity`)

This metric quantifies how closely the GLOBE segmentation resembles a perfect sphere. Deviations can indicate distortions, incomplete reconstruction, or other artifacts. The "sphericity" IQM quantifies the similarity of the GLOBE segmentation to a perfect sphere. Using the `skimage` library, we can obtain the volume (V) and the surface area (A). The sphericity index (S), as defined by [12], is a dimensionless quantity that describes how closely a given object resembles a perfect sphere.

$$S = \frac{\pi^{1/3}(6V)^{2/3}}{A} \quad (3.1)$$

Here, V represents the volume and A is the surface area of the object. A value close to 1 means that the segmentation is close to a perfect sphere, a 0 means a perfect plane

Lens Aspect Ratio (`seg_lens_aspect_ratio`)

This metric characterizes the shape of the lens, in particular its elongation, thanks to the ratio between its major axis and its minor axis. This IQM is useful because a good segmentation of the lens is generally ellipsoidal in shape, while a bad segmentation of the lens has more of a sphere shape.

We use `skimage.measure.regionprops_table` to obtain the eigenvalues of the inertia tensor. The lens aspect ratio, calculated as the ratio of the square root of the minimum to the maximum eigenvalues of the second moment matrix.

$$AspectRatio = \frac{\sqrt{\min(\lambda_i)}}{\sqrt{\max(\lambda_i)}}. \quad (3.2)$$

Here, λ_i represent the eigenvalues of the object's moment of inertia tensor. A value close to 1 means that the segmentation is close to a perfect sphere, a 0 means a perfect plane

3.5 Systematic Application of Segmentation-Based Statistics and Topology to All Eye Tissues

The FetMRQC_SR framework's capability to derive statistics for any segmented region was extended to all defined ocular tissues (LENS, GLOBE, OPTIC_NERVE, FAT, MUSCLE).

- **General Statistics** (`seg_sstats_[TISSUE]_*`, `seg_volume_[TISSUE]_*`, `seg_snr_[TISSUE]_*`): The wrappers for calculating summary statistics (mean, median, std, percentiles, etc.), volume fractions, and tissue-specific Signal-to-Noise Ratios is adapted to iterate over all classes.
- **Topological Metrics** (`seg_topology_[TISSUE]_*`, `seg_topology_mask_*`): The wrapper for topological metrics was similarly updated to process each individual eye tissue mask, as well as a combined mask of all eye tissues. It computes Betti numbers ($\beta_0, \beta_1, \beta_2$) and the Euler characteristic. This allows for the assessment of the structural integrity and connectivity of each segmented ocular component.

3.5.1 Adaptation of Segmentation-Based Statistics

The generic IQM (`seg`) using segmentation, `seg_sstats`, `seg_volume`, `seg_snr`, `seg_topology`, `betty` (`b0`, `b1`, `b2`) have been adapted to use the defined classes of eyes (LENS, GLOBE, OPTIC_NERVE, FAT, and MUSCLE.). You can see in the appendix which class was used for each IQM.

4 | Results and Current Status

This chapter presents current results of the MREyeQC_PA project, focusing on the successful adaptation of the Image Quality Metric (IQM) computational pipeline, and the first results of learning-based quality prediction models.

4.1 IQM Computation Pipeline for Ocular MRI: Validation and Output

With the modifications to the FetMRQC_SR project, the pipeline for computing all IQMs for ocular MRI is now operational. At the end of this pipeline, a CSV file is obtained that aggregates the results of each IQM for each subject tested. This dataset constitutes the direct input to the machine learning workflow for automated quality assessment.

4.2 Correlation index of each IQM

Analyzing the calculated IQM file manually proves to be complicated. Therefore, the correlation index of each IQM according to the desired result was calculated to observe which IQMs are most relevant for classifying the images. The result obtained is presented below.

Table 4.1 Top 20 IQMs Correlated with 'Rating' (Absolute Values)

IQM	Correlation Coefficient
seg_snr_GLOBE	0.682922
seg_cjv	-0.626543
seg_sstats_GLOBE_mad	-0.573834
seg_sstats_GLOBE_stdv	-0.550379
seg_sstats_GLOBE_k	0.485470
seg_snr_total	0.469745
seg_cnr	0.468125
seg_sstats_GLOBE_p95	-0.445921
centroid_full	0.359771
seg_sstats_MUSCLE_k	0.355294
seg_sstats_OPTIQUE_NERVE_n	0.282753
mi_median	-0.262807
seg_sstats_FAT_mad	-0.259243
shannon_entropy	-0.258915
mi_window	-0.253138
seg_sstats_GLOBE_p05	0.240962
seg_lens_aspect_ratio	-0.236386
seg_sstats_FAT_stdv	-0.236090
seg_sstats_OPTIQUE_NERVE_p95	-0.232774
nmi_median	-0.225741

4.2.1 Observation

First, it is observed that none of the IQMs achieved a very high correlation coefficient, with the maximum absolute correlation being 0.683 for seg_snr_GLOBE. This indicates that while useful, no single IQM perfectly aligns with human quality ratings in this dataset.

Notably, IQMs derived from the globe segmentation (GLOBE class) consistently show strong correlations with the ratings. In fact, among the top 8 correlated IQMs, all are based on segmentation features. This strongly suggests that the quality of anatomical segmentation is a crucial factor for accurately classifying image quality.

Conversely, it is important to highlight that none of the newly implemented IQMs appeared in the top 20 correlations. This suggests that initial hypotheses regarding these specific new metrics were not supported by the data, and they may not be effective indicators for image quality classification in this context.

4.3 Machine Learning for Automated Quality Prediction: Initial Model Performance

The machine learning phase aimed to develop a model capable of predicting image quality from the calculated image quality indicators (IQMs). This section details the characteristics of the dataset, the model training method, and preliminary performance results. As illustrated in Figure 4.1, the distribution of the original multi-category ratings (Exclude, Poor, Acceptable, Excellent) highlights a non-homogeneous representation across classes. For the purpose of binary classification, these ratings were consolidated into "reject" (corresponding to the "excluded" category) and "accept" (encompassing "poor," "acceptable," and "excellent"). The resulting binary distribution, shown in Figure 4.1, underscores a class imbalance, which necessitated specific considerations during model training. The relatively small total dataset size ($N=83$) inherently presents challenges for model generalization and amplifies the sensitivity of performance metrics to data splits.

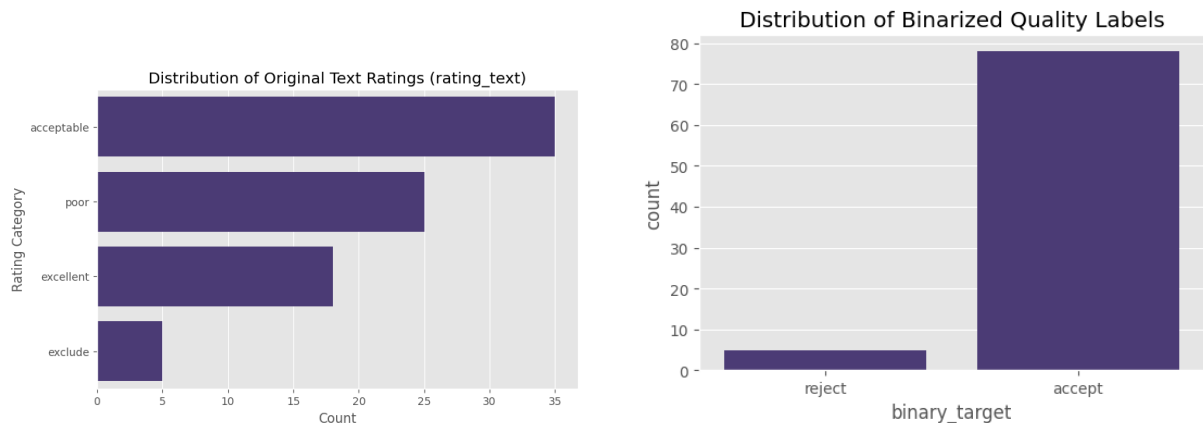


Figure 4.1 Comparison of manual image quality rating distributions: (a) Original multi-category ratings (Exclude, Poor, Acceptable, Excellent) across 83 images, and (b) Binarized quality labels ("reject" vs. "accept") used for model training.

A Random Forest classifier was chosen as the training model. The dataset was subsequently partitioned into training (75%) and testing (25%) sets using stratified sampling to maintain class proportions across both splits. Standard preprocessing involved feature scaling via StandardScaler. To address class imbalance, the Random Forest model was configured with `class_weight='balanced'`. Hyperparameter tuning was performed using 5-fold stratified cross-validation on the training set, with ROC AUC as the optimization metric. The final model's optimized parameters were: `n_estimators=100`, `max_depth=20`, `min_samples_split=5`, and `min_samples_leaf=2`.

4.3.1 Prediction Generation and Preliminary Model Efficacy

The optimized Random Forest classifier was subsequently utilized to generate automated quality predictions on the held-out test set. This test set comprised 20 images, consisting of 3 'reject' cases (15.0% of the test set) and 17 'accept' cases (85.0% of the test set)

Table 4.2 Per-Class Performance Metrics on the Test Set for the Random Forest Model

Class	Precision	Recall (Sensitivity)	F1-score
Reject (Class 0)	0.00	0.00	1
Accept (Class 1)	0.95	1.0	0.98

The confusion matrix for the test set and The Receiver Operating Characteristic (ROC) curve are presented in Figure 4.2.

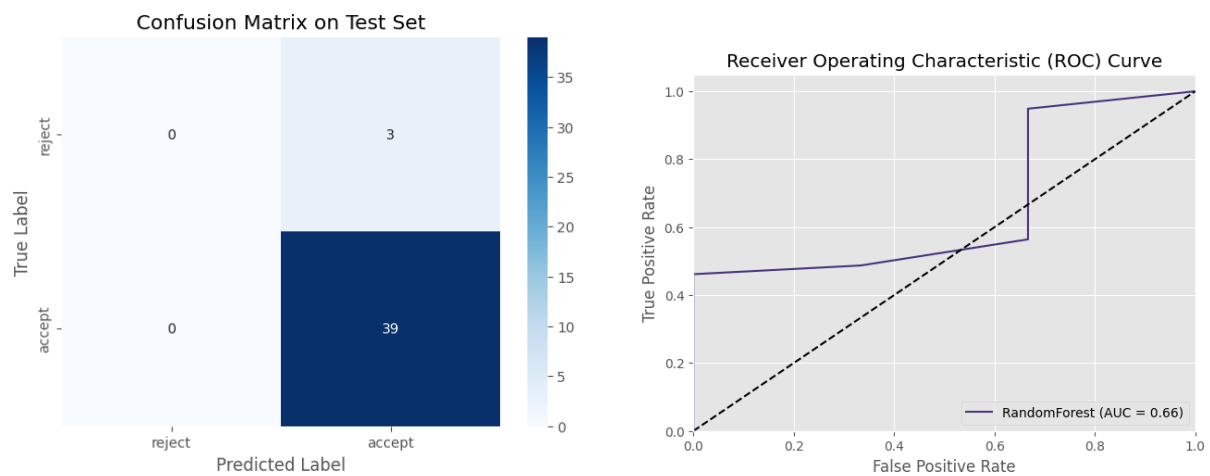


Figure 4.2 Performance of the Random Forest model on the test set: (a) Confusion matrix, illustrating true/false positives and negatives for the "accept" and "reject" classes; and (b) Receiver Operating Characteristic (ROC) curve with an Area Under the Curve (AUC) of 0.6581.

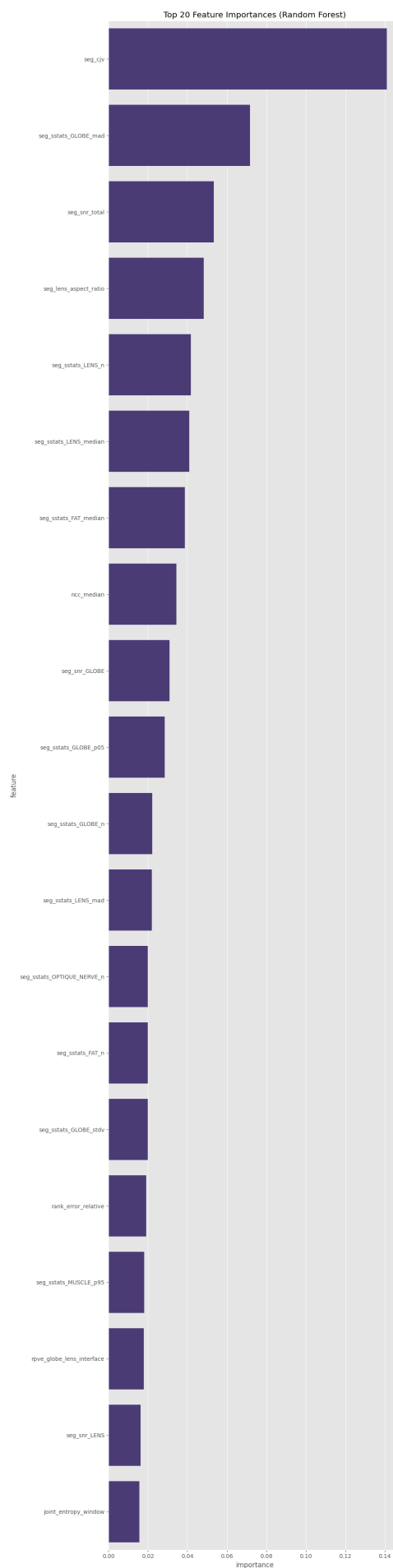


Figure 4.3 Top N feature importances from the Random Forest model, indicating the relative contribution of different IQMs to the quality prediction.

Interpretation of Preliminary Results: The reported overall accuracy of 0.9286 appears high. However, the ROC AUC of 0.6581 suggests a modest discriminative ability of the model across all decision thresholds. Given the imbalanced nature of the test set, the high overall accuracy and the reported recall of 1.0 and F1-score of 0.9630 are likely dominated by the model's performance on the majority "accept" class.

The critical evaluation hinges on the Balanced Accuracy and the recall for the "reject" class. A Balanced Accuracy significantly lower than the overall accuracy would confirm the influence of class imbalance. For instance, if the recall for the "reject" class is low, it would mean that while most good quality images are correctly classified, a substantial portion of poor quality ("excluded") images are being missed (false negatives for the "reject" class). Conversely, the precision for the "reject" class would indicate the rate of false alarms.

These initial findings, while demonstrating the feasibility of training a predictive model, highlight the challenges posed by the small dataset size and class imbalance. The current ROC AUC suggests room for improvement in the model's overall ability to separate the classes robustly. Future work will focus on.

5 | Conclusion and Future Perspectives

This project embarked on the significant challenge of adapting and extending the FetMRQC_SR framework, a tool originally developed for fetal brain MRI, to establish a robust pipeline for the automated Quality Control (QC) of Super-Resolution Reconstructed ocular MRI. The primary objective was to develop a system capable of quantitatively assessing eye MR image quality through a tailored suite of Image Quality Metrics (IQMs) and subsequently to leverage these metrics for training machine learning models for automated quality prediction.

The project successfully achieved its core aims. Key accomplishments include the meticulous re-targeting of the segmentation data management to accurately process defined ocular anatomical classes (LENS, GLOBE, OPTIC_NERVE, FAT, MUSCLE), incorporating essential preprocessing steps such as robust label mapping and right-eye spatial filtering. The IQM suite itself was substantially enhanced: existing general-purpose and brain-specific metrics (e.g., CNR, CJV, SNR, Dietrich, FBER, EFC) were carefully adapted for ophthalmic application, and novel, eye-specific morphological IQMs – Globe Sphericity and Lens Aspect Ratio – were successfully designed and implemented to capture unique anatomical features. Furthermore, established MRIQC-inspired metrics (QI1, QI2, Tissue-to-Max ratio, RPVE) and comprehensive segmentation-based statistics, including topological features, were integrated, resulting in a rich, quantitative feature set extracted from eye MRI data. Throughout this adaptation, notable technical hurdles pertaining to data type compatibility, argument propagation across modules, handling outputs from external libraries, and ensuring morphological operation versioning were systematically addressed and overcome.

The adapted MREyeQC_PA system now capably and extracts this extensive set of quantitative features, outputting them into a structured `IQA.csv` file. This forms the essential input for the machine learning component of the QC pipeline. Preliminary analysis of these IQMs revealed their potential sensitivity to variations in image quality and specific artifacts, as indicated by correlation studies with manual ratings (Section 4.3).

This project successfully lays a crucial and comprehensive foundation for automated, objective quality assessment in the specialized domain of eye imaging. The developed pipeline and the extensive, tailored IQM set offer a significant step forward from subjective manual QC, paving the way for more reliable and reproducible quantitative studies in ophthalmic research. The systematic approach to adapting a brain-focused framework also provides valuable insights for similar cross-domain QC tool development.

Future work should prioritize several key areas. Firstly, dataset expansion is paramount, with a focus on acquiring a larger number of eye images and, crucially, enriching the representation of images with diverse quality levels, especially those manually rated as "excluded" or "poor." This will be essential for training more generalizable and robust machine learning models. Secondly, further model exploration and refinement are warranted, including experimenting with advanced feature selection techniques, alternative classification algorithms, and methods specifically designed for imbalanced learning beyond 'class_weight'. Thirdly, the heuristic definitions for complex IQMs such as the artifact indices (QI1, QI2) and RPVE should be further refined and validated, potentially through correlation with specific, visually identified artifacts or through more sophisticated mask generation techniques. Finally, prospective validation of the most promising automated QC model on unseen data from different scanners or sites would be an important step towards demonstrating its broader applicability and potential for clinical or research integration.

In conclusion, the MREyeQC_PA project has made substantial progress in addressing the need for automated quality control in ocular MRI. While the initial predictive models highlight areas for improvement, the successfully adapted IQM pipeline provides a robust and extensible platform for future advancements, ultimately aiming to enhance the quality and reliability of quantitative eye imaging research and its clinical applications.

A | Distribution by rater across



Figure A.1 Distribution of artifact severity ratings by rater across different categories: (a) Background Artifacts (Bgair), (b) Blur, (c) Motion, and (d) Noise. These pie charts illustrate variations in how frequently each rater assigned images to the various quality categories for each specific artifact.

B | QMs already implemented in FetMRQC_SR

I. Global Intensity Distribution & Volumetric Properties

These metrics describe the overall intensity characteristics and basic volume information within the primary region of interest.

- **mean**: Mean image intensity.
- **median**: Median image intensity.
- **std**: Standard deviation of image intensities.
- **variation**: Coefficient of variation (std/mean) of image intensities.
- **kurtosis**: Kurtosis of the image intensity distribution.
- **percentile_5 / percentile_95**: 5th and 95th percentiles of image intensity.
- **centroid**: Centroid of image intensities based on a defined mask.
- **centroid_full**: Centroid of image intensities, potentially based on a broader mask or the full image data.
- **mask_volume**: Volume of the primary analysis mask.

II. Information-Theoretic Measures

These metrics quantify the information content or complexity of the image intensity distribution.

- **shannon_entropy**: Shannon entropy of the image intensity distribution.
- **joint_entropy_median**: Joint entropy, typically calculated between adjacent image slices using median aggregation.
- **mi_window**: Mutual Information, typically calculated between adjacent image slices using windowed aggregation.
- **nmi_window / nmi_median**: Normalized Mutual Information, using windowed or median aggregation.

III. Inter-Slice / Window-Based Consistency & Image Quality

These metrics are often calculated between adjacent slices or within defined windows to assess consistency, noise, and structural similarity.

- **ncc_window / ncc_median**: Normalized Cross-Correlation.
- **psnr_window**: Peak Signal-to-Noise Ratio.
- **rmse_window / nrmse_window**: Root Mean Squared Error and its normalized version.
- **mae_window / nmae_window**: Mean Absolute Error and its normalized version.

- `ssim_window`: Structural Similarity Index.

IV. Edge Detection / Filter-Based Metrics

These metrics utilize image filters to highlight edges or textural features.

- `filter_laplace`: Result from applying a Laplacian filter (sensitive to edges/texture).
- `filter_sobel`: Result from applying a Sobel filter (edge detection).

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